

SIMATS ENGINEERING
CO3 - ASSESSMENT TOOL - 1: Experiments

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Shri Shanmuga Priya K
Register Number	192324035

COURSE OUTCOME COVEREDCO3: *Implement cloud-based solutions using cloud storage, web APIs, and platform services for real-world applications. (BL3)*

Assessment Tool Weightage: 40%

Dave's Taxonomy: Precision (Level 3)
Question Count: 4

Total Marks: 40

Duration : 60 Minutes

Code of Conduct

I Shri Shanmuga Priya K 192324035 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Experiments Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Experimental Design	25%	Design is systematic and technically strong	Mostly correct design	Basic design with gaps	Poorly designed activity
Use of Tools / Platforms	20%	Appropriate and effective use of cloud tools	Good tool usage with minor issues	Limited tool usage	Incorrect or missing tool usage
Application of Concepts	20%	Concepts are applied accurately to practical tasks	Mostly accurate application	Partial application	Weak application
Analysis and Output	20%	Strong analysis and meaningful outcome interpretation	Good analysis with minor gaps	Basic analysis	Little or no analysis
Documentation	15%	Clear, structured, and complete documentation	Mostly complete documentation	Partially complete	Poor documentation

Total Marks Awarded:

Assessment Tasks

Experiments

Experiment 1: Create a simple cloud software application for Library book reservation system for SIMATS library using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Title of the book, author, publications, Year, edition, no of copies, Rack no, status (taken/returned) etc.

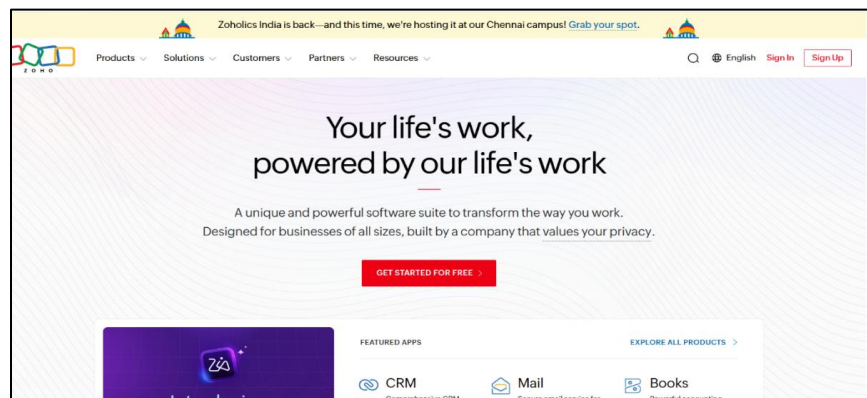
Aim: To create a cloud-based Library Book Reservation System using Zoho Creator.

Steps:

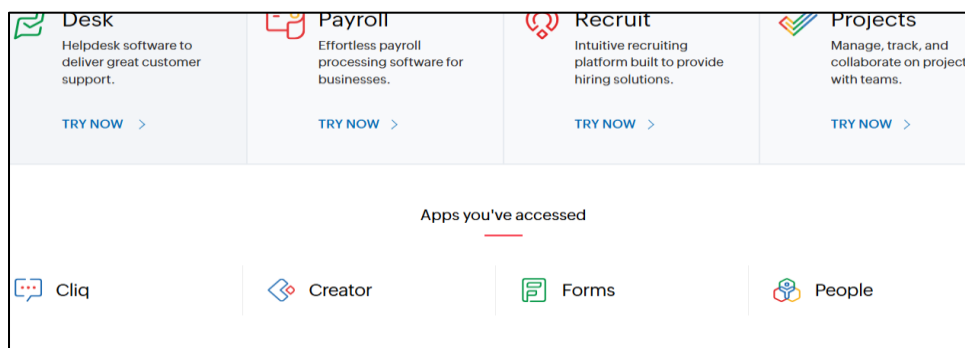
1. Create a new application.
2. Create a Book Reservation form.
3. Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status.
4. Create a report to display available and reserved books.
5. Test the application with sample book records.

Implemented

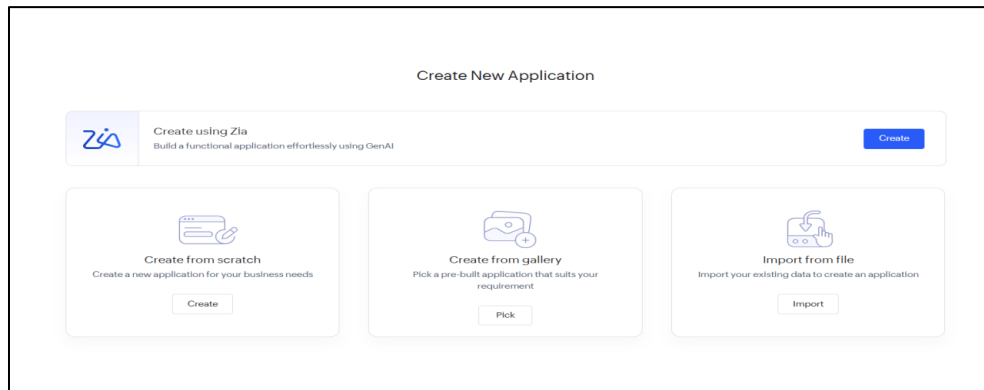
STEP 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



STEP 3: Click Create New Application and select Create from Scratch



STEP 4: Enter Library Book Reservation System as the application name and click Create Application.

The screenshot shows the 'Create from scratch' form. It has a title 'Create from scratch' with a logo above it. Below the title, there's a section for 'Application Name *' with a text input field containing 'Library Book Reservation System'. There's a checkbox labeled 'Enable Environment' with a link 'Learn More'. At the bottom, there's a large blue button labeled 'Create Application'.

Step 5: Add fields: Book Title, Author, Publication, Edition, Rack Number, Number of Copies, and Status

The screenshot shows the 'Library Management System' interface. On the left, there's a sidebar with a menu: 'Dashboard', 'Issue Records' (expanded), 'Issue Records Rep...', 'Board', 'Borrowing History', 'Books', 'Students', 'Return Records', 'Dashboard Metrics', and 'Items'. The main area is titled 'Issue Records' and contains a form with the following fields: 'Student' (text input), 'Book' (dropdown menu), 'Issue Date' (date input), 'Return Date' (date input), 'Due Date' (date input), 'Status' (dropdown menu), 'Fine Amount' (text input), and 'Returned By' (text input). At the bottom, there are 'Submit' and 'Reset' buttons.

SIMATS Library Management Sys...
Book Issue Return

Basic Fields

Name Email

Address Phone

Single Line Multi Line

123 Number 15 Date

Time Drop Down

Transaction Date

Issue Date

Due Date

Return Date

Fine Amount

Transaction Status

Transaction Code

Reservation

Student

Book

SIMATS Library Management System

SIMATS Library Manageme...

Dashboard

Book Issue Return

Book Reservations

Book Details

+ Book Details

Book Details Report

Book Categories

Student Registrati...

Fine Payments

Book Details

Book Title

Author

Publication

Publication Year

Edition

ISBN Number

Book Category

Number of Copies

Available Copies

Rack Number

Shelf Number

SIMATS Library Manageme...

Dashboard

Book Issue Return

Book Reservations

Book Details

Student Registrati...

+ Student Registration

All Student Registr...

Fine Payments

Student Registration

Student Name

First Name

Department

Course

Year

Section

Mobile Number

+91 81234 56789

Email Address

Address

Address Line 1

Registration Date

dd-MMM-yyyy

Student Code

STEP 6: Save the form, enter sample booking records, and view them in the Library Book Reservation System.

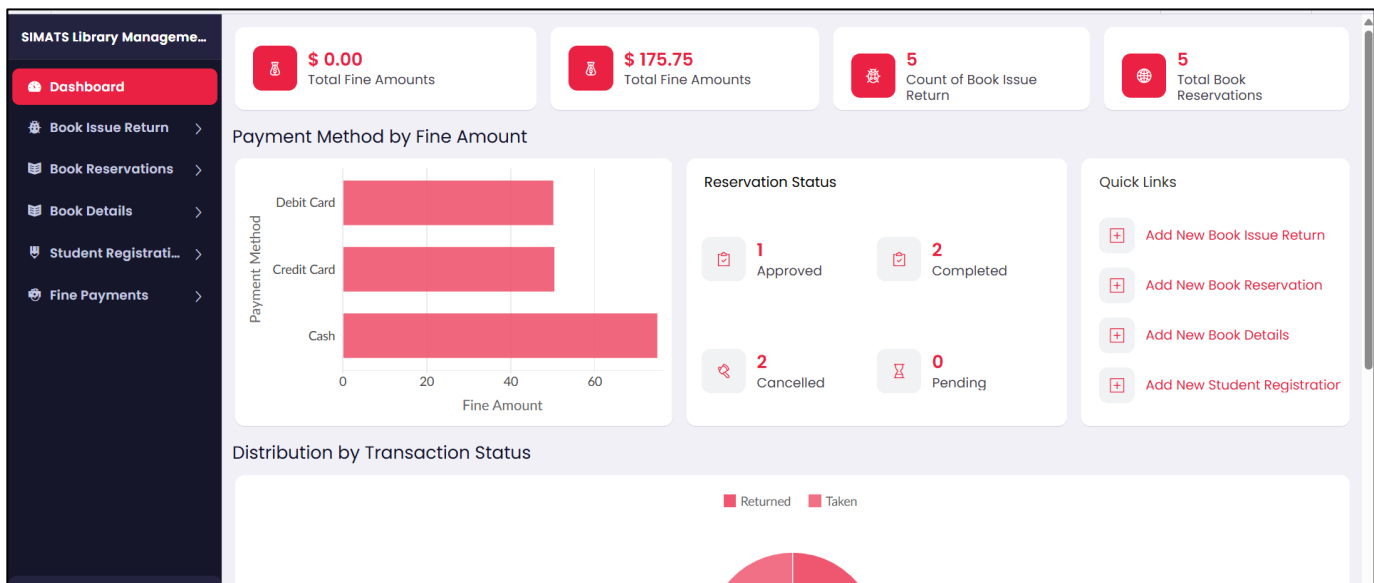
SIMATS Library Manageme... Book Issue Return Report

Transaction Date	Issue Date	Due Date	Return Date	Fine Amount	Transaction Status	Transaction Code	Reservation
11-Aug-2026	11-Aug-2026	18-Aug-2026	18-Aug-2026	\$ 0.00	Returned	TRX005	RES-2026-002
10-Aug-2026	10-Aug-2026	17-Aug-2026	17-Aug-2026	\$ 0.00	Taken	TRX004	RES-2026-001
09-Aug-2026	09-Aug-2026	16-Aug-2026	16-Aug-2026	\$ 0.00	Taken	TRX003	RES-2026-001
08-Aug-2026	08-Aug-2026	15-Aug-2026	15-Aug-2026	\$ 0.00	Taken	TRX002	RES-2026-004
07-Aug-2026	07-Aug-2026	14-Aug-2026	14-Aug-2026	\$ 0.00	Returned	TRX001	RES-2026-004

SIMATS Library Managemen...		All Student Registrations								
		Student Name	Department	Course	Year	Section	Mobile Number	Email Address	Address	Registr
		Robert Chen	Engineering	Software Design	2026	B	+19015986438	robertc@university.edu	505 Eng Blvd	07-Aug
		Maria Garcia	Biology	Cell Biology	2026	C	+18729908100	mariag@university.edu	404 Bio Way	07-Aug
		James Wilson	Mathematics	Calculus III	2026	A	+18298545159	jamesw@university.edu	303 Math Ln	07-Aug
		Sarah Connor	Physics	Quantum Mechanics	2026	B	+18192467340	sarahc@university.edu	202 Science Rd	07-Aug
		Alex Morgan	Computer Science	Data Structures	2026	A	+17022597323	alexm@university.edu	101 Campus Dr	07-Aug

SIMATS Library Managemen...		All Fine Payments							
		Payment Date	Payment Method	Payment Status	Payment Code	Student	Transaction	Fine Amount	
		07-Aug-2026	Cash	Paid	PAY-005	STU2026004	TRX001	\$ 75.00	
		07-Aug-2026	Credit Card	Pending	PAY-004	STU2026003	TRX002	\$ 10.00	
		07-Aug-2026	Debit Card	Paid	PAY-003	STU2026002	TRX004	\$ 50.25	
		07-Aug-2026	Credit Card	Paid	PAY-002	STU2026005	TRX004	\$ 15.00	
		07-Aug-2026	Credit Card	Paid	PAY-001	STU2026001	TRX005	\$ 25.50	

STEP 7: Final Dashboard.



Result: Successfully created a cloud-based Library Book Reservation System using Zoho Creator.

Experiment 2: Create a simple cloud software application for Payroll Processing System using any Cloud Service Provider to demonstrate SaaS include the necessary fields such as Name of the employee, experience in present org, over all experience, basic pay, HRA, DA, TA, net pay, gross pay etc.

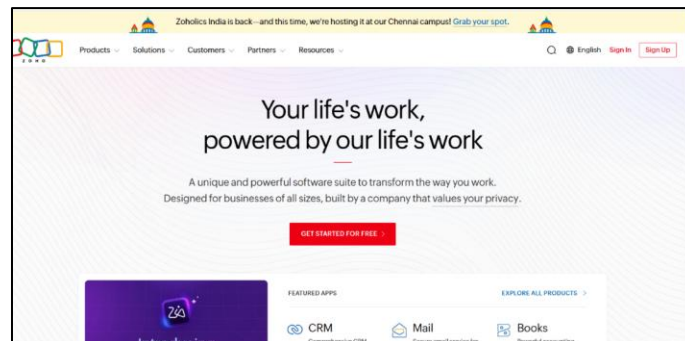
Aim: To create a cloud-based Payroll Processing System using Zoho Creator.

Steps:

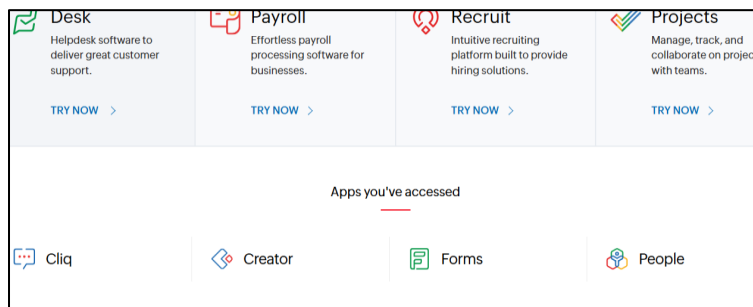
1. Create a new application.
2. Create an Employee Payroll form.
3. Add fields: Employee Name, Experience, Basic Pay, HRA, DA, TA, Gross Pay, and Net Pay.
4. Use formulas to calculate Gross Pay and Net Pay.
5. Save employee details and verify the payroll report.

Implemented:

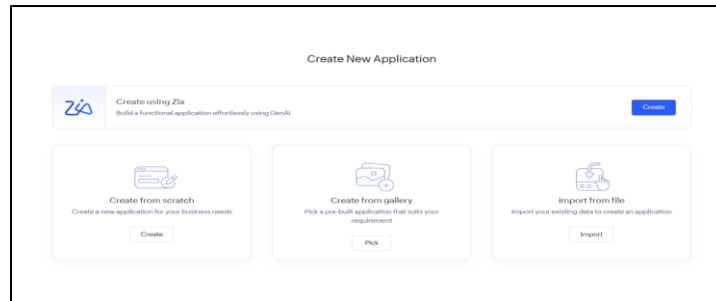
Step 1: Go to zoho.com



STEP 2: Login to your Zoho account and open Zoho Creator



STEP 3: Click Create New Application and select Create from Scratch



STEP 4: Enter Payroll Processing System as the application name and click Create Application.

The screenshot shows the 'Create from scratch' form. The 'Application Name' field is filled with 'Payroll Processing System'. Below it, there's a checkbox for 'Enable Environment' which is unchecked, and a link 'Learn More'. At the bottom, there's a large blue button labeled 'Create Application'.

Step 5: Add fields: Employee Name, Employee Code, Salary, Total, Average, Percentage, and PaySlip.

The screenshot shows the 'Employee Registration' form. It has a sidebar with navigation links: Dashboard, Employee Registr..., All Employee Regi..., Genders, Salary Structures, Monthly Payrolls, Payslip, Management, Zoho People Depa..., and Zoho People Desig... The main form fields are: Employee Name (text input), Employee Code (text input), Gender (dropdown), Date of Birth (dd-MM-yyyy), Department (dropdown), Designation (dropdown), Date of Joining (dd-MM-yyyy), Experience in Present Org (text input), Overall Experience (text input), Mobile Number (text input with a country code dropdown), and Email Address (text input).

The screenshot shows the 'Salary Structure' form. It has a sidebar with navigation links: Dashboard, Employee Registr..., Salary Structures, All Salary Structures, Monthly Payrolls, Payslip, Management, Zoho People Depa..., and Zoho People Desig... The main form fields are: Employee (text input), Basic Pay (text input), House Rent Allowance (text input), Dearness Allowance (text input), Travel Allowance (text input), Medical Allowance (text input), Special Allowance (text input), Bonus (text input), Provident Fund (text input), and Professional Tax (text input).

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Management

Zoho People Depa...

Zoho People Desig...

Monthly Payroll

Payroll Month

Payroll Year

Employee

Working Days

Days Present

Leave Days

Overtime Hours

Overtime Amount

Gross Pay

Total Deductions

Net Pay

STEP 6: Save the form Payroll Processing, enter sample booking records, and view them in the System.

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Management

Zoho People Depa...

Zoho People Desig...

All Monthly Payrolls

Payroll ID	Payroll Month	Payroll Year	Employee	Working Days	Days Present	Leave Days	Overtime Hours	Overtime A
5	April	2026	Robert Williams Sarah Martinez Michael Chen Jennifer Taylor	22	18	4	6	\$
4	May	2026	Alice Johnson	22	22	0	4	\$
3	June	2026	Alice Johnson Robert Williams Michael Chen	22	19	3	6	\$
2	July	2026	Alice Johnson	22	21	1	5	\$
1	August	2026	Alice Johnson Robert Williams Sarah Martinez	22	20	2	4	\$

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Payslip Report

Management

Zoho People Depa...

Zoho People Desig...

Payslip Report

Employee Name	Employee Code	Department	Designation	Payroll Month	Payroll Year	Basic Pay	HRA	DA
Arnit Verma	EMP005	Ops	Operations Head	Aug	2026	\$ 60,000.00	\$ 12,000.00	\$ 6,000.00
Priya Patel	EMP004	Sales	Sales Executive	Aug	2026	\$ 35,000.00	\$ 7,000.00	\$ 3,500.00
Vikram Singh	EMP003	Finance	Accountant	Aug	2026	\$ 40,000.00	\$ 8,000.00	\$ 4,000.00
Anita Sharma	EMP002	HR	HR Manager	Aug	2026	\$ 45,000.00	\$ 9,000.00	\$ 4,500.00
Rajesh Kumar	EMP001	IT	Senior Developer	Aug	2026	\$ 50,000.00	\$ 10,000.00	\$ 5,000.00

Cloud Payroll Processing S...

Dashboard

Employee Registr...

Salary Structures

Monthly Payrolls

Monthly Payroll

All Monthly Payrolls

Statuses

Payslip

Management

All Management

Leave Management

Leave Types

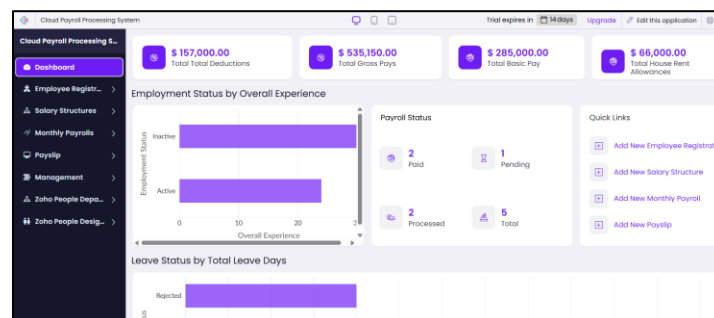
Zoho People Depa...

Zoho People Desig...

All Management

Employee	Leave Type	From Date	To Date	Total Leave Days	Leave Status
Alice Johnson Robert Williams Sarah Martinez Michael Chen Jennifer Taylor	Unpaid Leave	21-Aug-2026	25-Aug-2026	4	Pending
Alice Johnson Robert Williams	Sick Leave	18-Aug-2026	20-Aug-2026	5	Approved
Alice Johnson Robert Williams	Unpaid Leave	11-Aug-2026	15-Aug-2026	4	Rejected
Alice Johnson Robert Williams Sarah Martinez	Sick Leave	06-Aug-2026	10-Aug-2026	5	Pending
Alice Johnson Robert Williams Sarah Martinez Michael Chen Jennifer Taylor	Casual Leave	01-Aug-2026	05-Aug-2026	3	Pending

STEP 7: Final Dashboard



Experiment 3: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Virtual Machine with 1 CPU, 2GB RAM and 15GB storage disk using a Type 2 Virtualization Software.

Aim: To create a virtual machine with 1 cpu, 2gb ram and 15gb storage disk using a type 2 virtualization software.

Procedure:

STEP 1: Download VMware workstation and installed as type 2hypervisor.

STEP 2: Download ubuntu or tiny OS as iso image file.

STEP 3: In VMware workstation->create new VM.

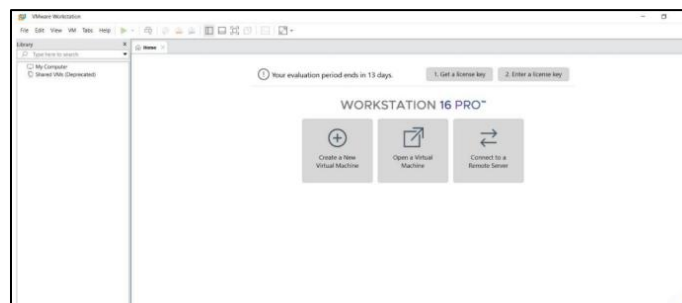
STEP 4: Do the basic configuration settings.

STEP 5: Created tiny OS virtual machine.

STEP 6: Launch the VM.

Implemenation:

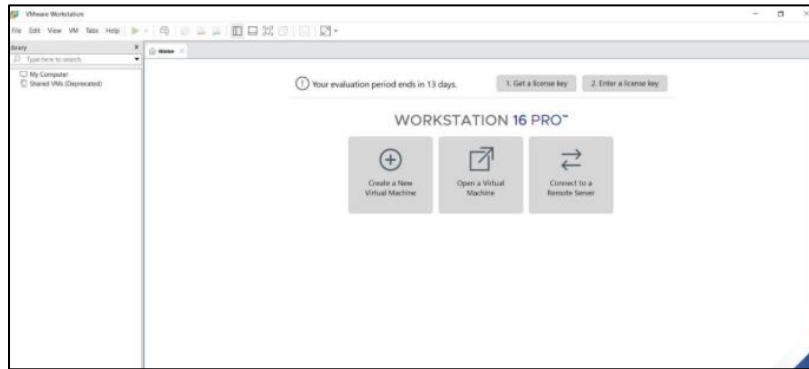
STEP 1: DOWLOAD VMWARE WORKSTATIONAND INSTALLED AS TYPE 2 HYPERVISOR



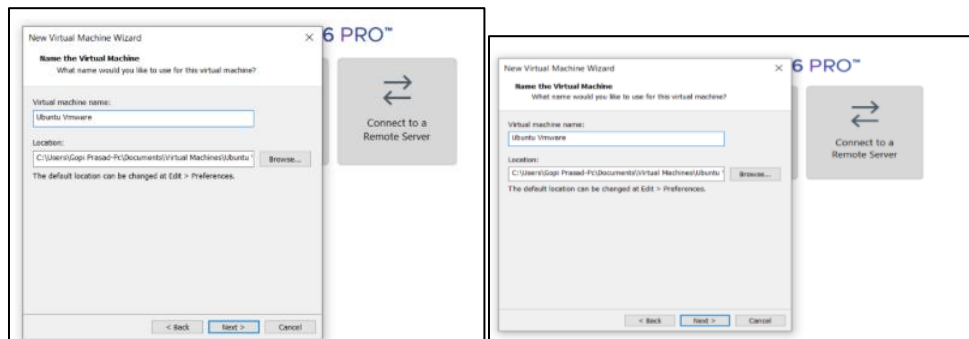
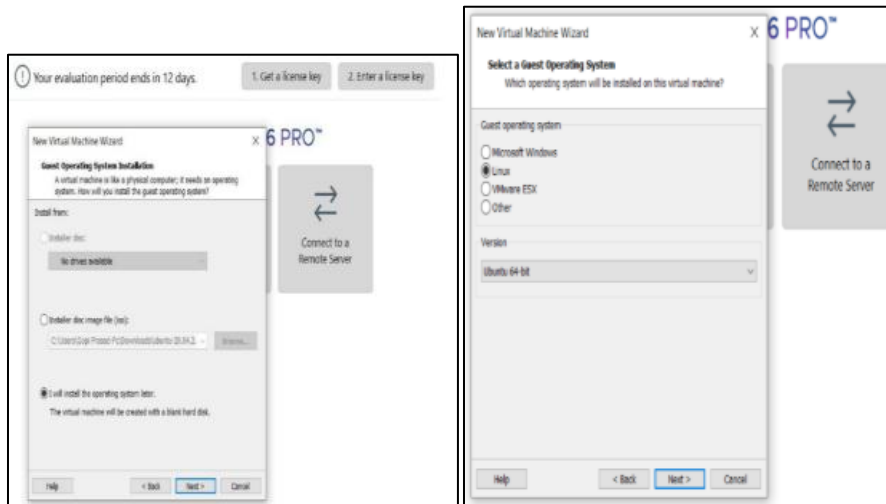
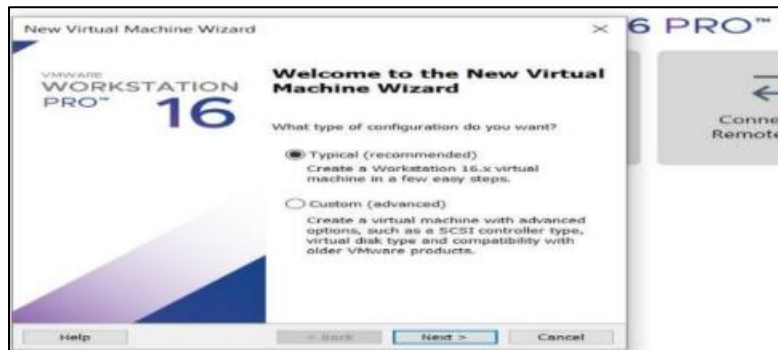
STEP2: DOWLOAD UBUNTU OR TINY OS AS ISO IMAGE FILE

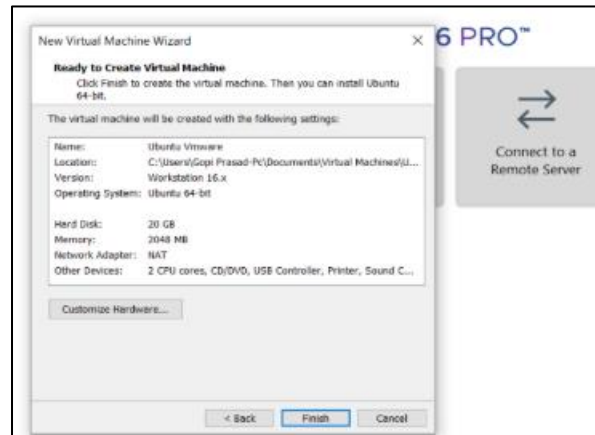
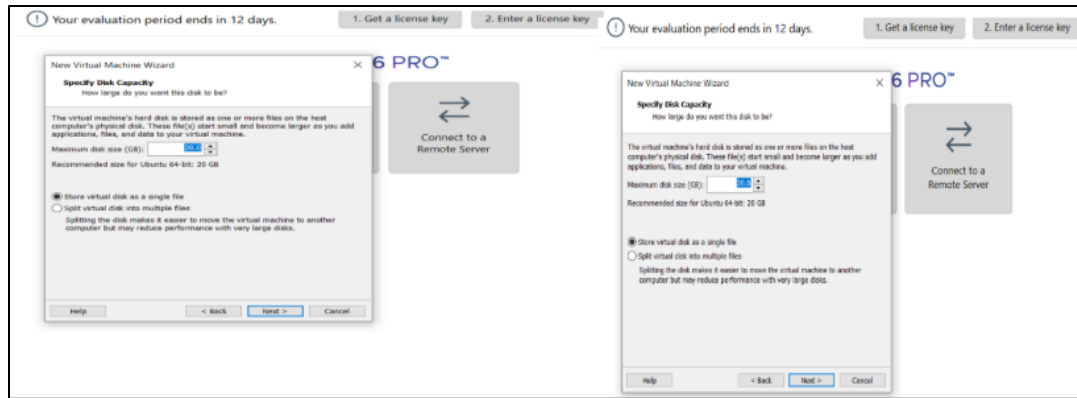
Index of /11.x/x86/release/		
..		
distribution_files/	09-Feb-2020 11:50	-
src/	03-Dec-2019 11:14	-
Core-11.1.iso	01-Apr-2020 07:49	14757888
Core-11.1.iso.md5.txt	01-Apr-2020 07:49	48
Core-11.1.iso.xsync	01-Apr-2020 07:49	50639
Core-current.iso	01-Apr-2020 07:49	14757888
CorePlus-11.1.iso	01-Apr-2020 07:50	216006656
CorePlus-11.1.iso.md5.txt	01-Apr-2020 07:50	52
CorePlus-11.1.iso.xsync	01-Apr-2020 07:50	369358
CorePlus-current.iso	01-Apr-2020 07:50	216006656
TinyCore-11.1.iso	01-Apr-2020 07:50	19922944
TinyCore-11.1.iso.md5.txt	01-Apr-2020 07:50	52
TinyCore-11.1.iso.xsync	01-Apr-2020 07:50	68301
TinyCore-current.iso	01-Apr-2020 07:50	19922944

STEP 3: IN VMWARE WORKSTATION->CREATE NEW VM

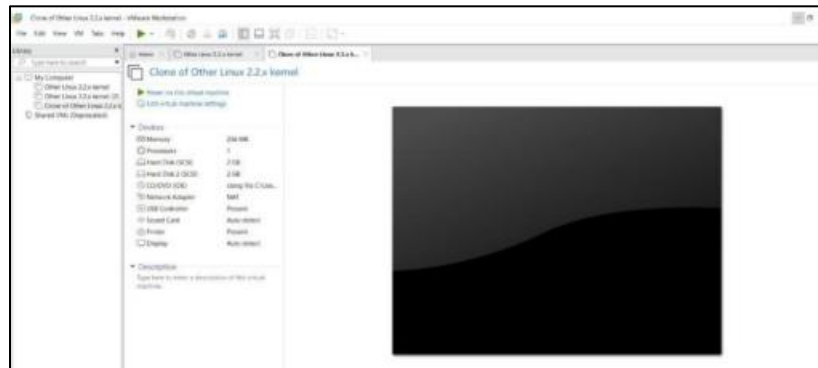


STEP 4: DO THE BASIC CONFIGURATION SETTINGS.





STEP 5: CREATED TINYOS VIRTUAL MACHINE



Result: Successfully created a virtual machine with 1 cpu, 2gb ram and 15gbstorage disk using a type 2 virtualization software.

Experiment 4: Demonstrate virtualization by Installing Type-2 Hypervisor in your device, create and configure VM image with a Host Operating system (Either Windows/Linux). Create a Cloning of a VM and Test it by loading the Previous Version/Cloned VM

Aim: To create a cloning of a vm and test it by loading the previous version/cloned vm.

Procedure:

STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE

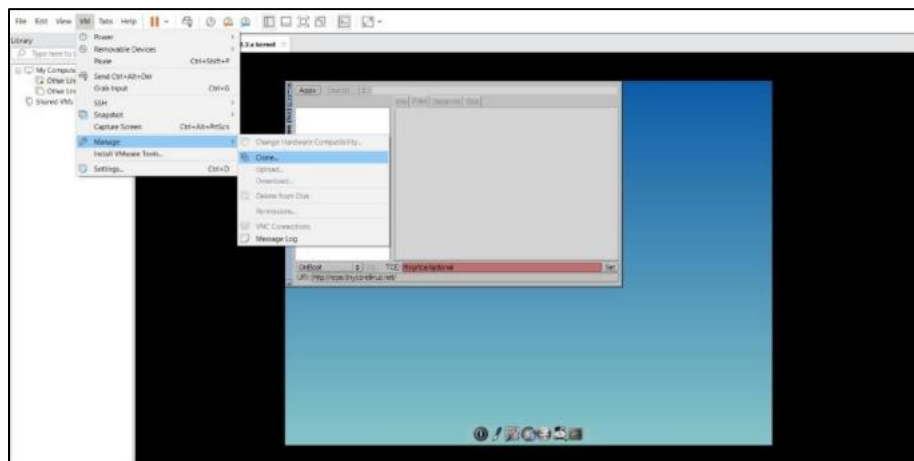
STEP 2: CLICK CLONE

STEP 3: SELECT THE FULL CLONE

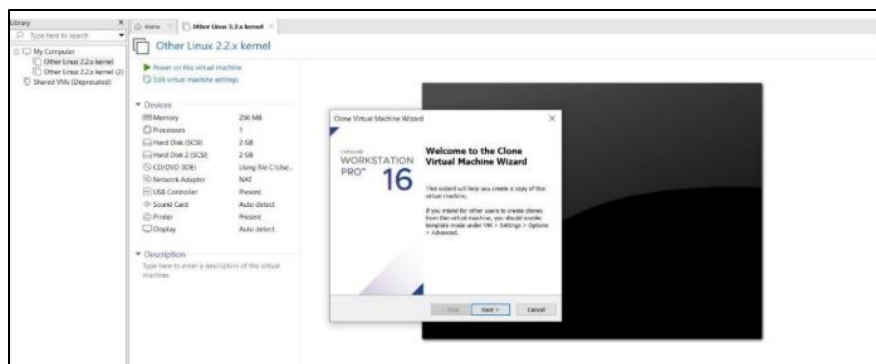
STEP 4: AFTER CLONE AGAIN OR VM IS OPENED.

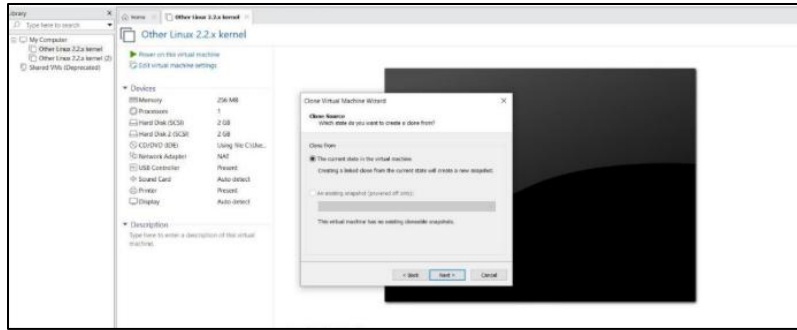
Implementation: CLONING OF A VM

STEP 1: GO TO VM AND GOTO MANAGE AND CLICK CLONE

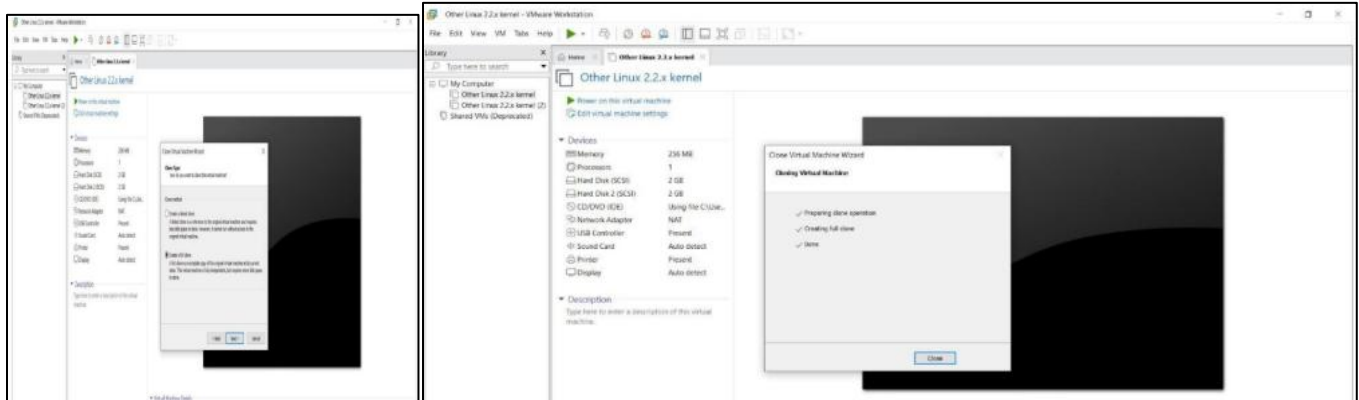


STEP 2: CLICK CLONE





STEP 3: SELECT THE FULL CLONE



STEP 4: AFTER CLONE AGAIN OR VM IS OPENED



Result: Successfully created a cloning of a vm and test it by loading the previous version/cloned vm.